

TASK-1

Networking Devices in a Home Wi-Fi Setup

Device	Main Function	Typical Ports / Interfaces	Example in a Home Network
Switch	Connects multiple wired devices and forwards data to the correct device using MAC addresses.	4–48 Ethernet (RJ-45) ports (1 Gbps or higher)	Connects desktop PCs, Smart TV, gaming console, and network printer.
Router	Connects the home network to the Internet and routes data between different networks. Also performs NAT and DHCP.	1 WAN (Internet) port, 4 LAN Ethernet ports , Wi-Fi antennas	Connects the home to the ISP and provides Internet access to all devices.
Hub	Connects multiple devices but sends incoming data to all connected ports without filtering.	4–24 Ethernet (RJ-45) ports	Rarely used today; suitable only for basic or legacy networking.
Firewall	Monitors and filters incoming and outgoing network traffic to improve security.	Usually no dedicated physical ports ; built into the router or a separate firewall appliance with WAN/LAN ports	Blocks unauthorized access while allowing safe Internet traffic.
Access Point (AP)	Provides wireless (Wi-Fi) connectivity to phones, laptops, tablets, and other wireless devices.	1 Ethernet uplink (RJ-45) and Wi-Fi radios (2.4 GHz / 5 GHz)	Extends Wi-Fi coverage to areas where the router's signal is weak.

TASK-2



Wireless Router:

- **Antennas:** Transmit/receive wireless Wi-Fi signals.
- **WAN (Internet) Port:** Connects to the modem/ISP line.
- **LAN (Ethernet) Ports:** Connect local wired devices (PCs, laptops, printers).
- **Power Port:** Plugs into the wall outlet via power adapter.



Managed Network Switch:

- **Ethernet Ports (RJ-45):** High-density ports (24 or 48) to connect multiple host devices.
- **Uplink / SFP Ports:** High-speed connections back to core routers.
- **Console Port:** Allows configuration via terminal/CLI using a console cable.
- **Power Input:** Standard AC power socket on the rear.



Enterprise Router:

- **Gigabit Ethernet (WAN/LAN) Ports:** Handle routing between distinct network segments.
- **Console Port (RJ-45 / Serial):** Direct terminal access for administrative access.
- **Antenna Connectors (SMA):** For cellular or Wi-Fi extension cards.
- **Power Switch & Port:** Main AC input unit.

TASK-3

Example: Instagram

When I open **Instagram** on my phone, my device first connects to the **Access Point (Wi-Fi router)**, which provides wireless Internet access. The **Router** then forwards my request from the home network to Instagram's servers over the Internet. A **Firewall** checks and filters incoming and outgoing traffic to help protect my network from unauthorized access. If I am using a wired network, a **Switch** connects multiple devices, such as my computer and smart TV, so they can communicate efficiently on the same local network.

TASK-4

Suggested Reason and Solution

One possible reason the **college Wi-Fi is slow and frequently disconnects** is that the **access point is overloaded** because too many students are connected to it at the same time. This can reduce Wi-Fi speed, increase delays, and cause devices to disconnect.

A good solution is to **upgrade the access point** to a newer model (such as Wi-Fi 6 or Wi-Fi 6E) that supports more simultaneous users and higher data speeds. Installing additional access points in crowded areas can also improve coverage and reduce congestion.